

Muscle Activity Detection Using Photoplethysmography and Machine Learning Algorithms: a Preliminary Study

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Abstract—This study evaluates the effectiveness of the MAX30102 sensor in detecting muscle activity and its potential for classifying hand movements. The sensor's working principle is based on photoplethysmography (PPG), which measures light absorption by blood in tissues and vessels. When light is projected through the skin, it encounters reflection from the flowing blood beneath the surface. The sensor employs photodiodes to measure the amount of light reflected by blood. Data collection was performed in the upper extremities of healthy users, with future validation intended for amputee patients. The study focused on data acquisition and preparation, followed by training machine learning algorithms to classify hand opening and closing movements. The results show that the Decision Tree and K-Nearest Neighbors models achieved high accuracy rates, with K-Nearest Neighbors reaching 98% and Decision Tree achieving 96%.

Index Terms—Photoplethysmography, muscle activity detection, machine learning, classification.

I. INTRODUCTION

In 2023, Brazil registered 645 hand and wrist amputation or disarticulation procedures [1]. Prostheses are essential tools that help restore autonomy and functionality to individuals with amputations or disabilities. The performance of these devices depends heavily on how accurately they can interpret the intended movements of the user. Accurate detection of these movement patterns is vital for the effective operation of underactuated prostheses, enabling users to perform everyday activities more naturally and confidently [2]. Hand movements are essential for approximately 85% of activities of daily living (ADLs) [3], involving various types of grips: power grip (35%), precision grip (30%), and lateral grip (20%) [4].

Understanding and measuring muscle activity is crucial in numerous fields, including human-computer interfaces, sports, rehabilitation, and prosthetics [5]. Muscle activity refers to the contraction and relaxation of muscle fibers, driven by electrical impulses originating from the central nervous system. These impulses trigger a cascade of chemical and mechanical processes within muscle fibers, resulting in static conditions,

where the muscle length does not change, or dynamic conditions, where the muscle length varies [6].

One of the most common methods for measuring muscle activity is electromyography (EMG), which records the electrical signals generated during muscle contractions using surface or needle electrodes [7]. Despite its widespread use, EMG has several limitations. It is susceptible to electromagnetic interference, which can distort signals and affect the accuracy of measurements [8]. Furthermore, achieving low skin electrode impedance requires proper skin preparation, such as cleaning the skin with alcohol, which can be cumbersome and time consuming [9]. Furthermore, the complexity of the electronics required to process the acquired signals can be a significant drawback, necessitating sophisticated equipment and expertise [10].

To address these limitations, this study proposes the use of photoplethysmography (PPG) as an alternative method. PPG is a non invasive optical technique that detects volumetric changes in blood in the peripheral circulation. It uses the light absorption properties of blood in tissues and vessels to detect physiological changes [11]. PPG offers several advantages over EMG. It is less susceptible to electromagnetic interference and eliminates issues with skin-electrode impedance by choosing appropriate wavelengths that penetrate tissues effectively [12]. Moreover, PPG uses simpler electronics, making it more accessible and easier to use in various settings [13].

Previous studies have demonstrated the potential of PPG for biomedical applications, including muscle activity detection [14] and the recognition of nine different hand gestures with an average accuracy of 89% [15]. However, no studies were found applying PPG for real-time control of prosthetic devices. This contrasts with Near-Infrared Spectroscopy (NIRS), a closely related optical technique, which has already been used in several studies [16]–[19].

Building on the foundation laid by previous research [20], this study integrates machine learning algorithms to enhance the characterization of muscle movements. The primary objective is to use the MAX30102 optical sensor data to classify hand opening and closing movements, with the goal of developing a system for the prosthetic control of underactuated devices [21].

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II. METHODOLOGY

This section describes the methodology used in the experiment, detailing data collection protocols, filtering techniques, feature extraction processes, and model training and evaluation methods.

A. Materials and Methods

The experiment used two types of sensors to capture and analyze muscle activity and hand movements. The Shimmer3 EMG sensor (Shimmer, Ireland) measures electrical activity in muscles and communicates wirelessly with a PC via Bluetooth. This setup enables real-time monitoring of electromyographic signals, capturing the electrical impulses generated by muscle contractions to provide direct measurements of muscle activity.

The MAX30102 optical sensor (Maxim Integrated, now part of Analog Devices, USA) employs red and infrared light-emitting diodes (LEDs) alongside a photonic filter to detect changes in blood volume. The emitted light penetrates the skin, where it is partially absorbed by the blood in tissues and vessels. The reflected light is then captured by photodiodes and converted into electrical signals. This technique provides insight into variations in blood volume associated with muscle activity. The sensor connects to a microcontroller (Arduino Uno, Arduino, Italy) via the I2C protocol and interfaces with a PC through a USB connection. Fig. 1 illustrates the experimental setup and its connections.

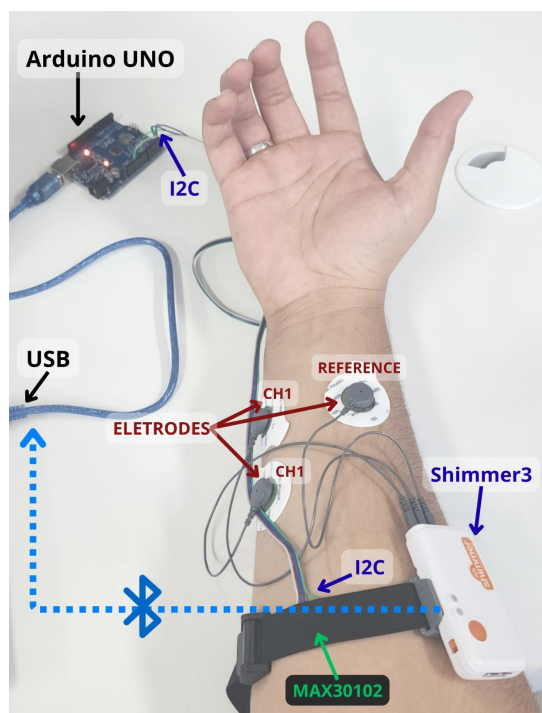


Fig. 1. Experimental setup showing the sensors' placement.

B. Data Collection Protocol

Data were collected from 20 healthy individuals, comprising 10 men and 10 women, with ages ranging from 18 to 50

for men and 18 to 67 for women. Data collection sessions occurred on different days and times to ensure diversity and robustness.

The sensors were placed on the right forearm over the Flexor Carpi Ulnaris muscle, ensuring that the arm remained still during the recording process. Data were recorded for 30 seconds, with the state of the hand alternating between open and closed every 3 seconds, a methodology commonly used in gesture recognition studies to facilitate segmentation and post-processing [22]. The objective was to classify the hand state using data obtained from the red and infrared LEDs of the MAX30102 sensor.

C. Data Filtering

During the experiment, a visual command was displayed on a computer screen every 3 seconds, instructing the participant to open or close their hand. However, data from initial attempts were excluded due to observed reaction delays in responding to these commands. These delays could compromise the accuracy of the initial data.

This preprocessing step was essential to remove unreliable data and mitigate bias in the training process, ensuring that the data set accurately reflected the responses of natural subjects. The remaining data were processed using a Butterworth low-pass filter to eliminate noise. The filter had a cutoff frequency of 0.8 Hz and an order of 4, determined by graphical analyzes. The results are presented on Fig. 2.

D. Feature Extraction

Feature extraction focused exclusively on calculating moving averages of the red and infrared signals. This choice was motivated by the requirement for real-time processing on resource-constrained hardware, where minimizing computational complexity is essential. The moving averages provided a smoothed representation of the sensor data, capturing the general trend of hand movements without the overhead of more complex calculations.

E. Model Training

Before training, the dataset was normalized using the `StandardScaler` from `scikit-learn`, which scales the data to have a mean of 0 and a standard deviation of 1, ensuring consistency across features.

The dataset was subsequently divided into training and testing subsets, with 80% used for training and 20% reserved for testing. Based on recommendations from [23], lightweight machine learning models suitable for microcontroller deployment were selected, including Logistic Regression, K-Nearest Neighbors, Decision Tree and Naive Bayes classifiers. These models were chosen for their low computational complexity and suitability for binary classification tasks in resource-constrained environments.

Hyperparameter optimization was performed using GridSearchCV from scikit-learn, which systematically explores predefined parameter grids to identify the best performing model configurations.

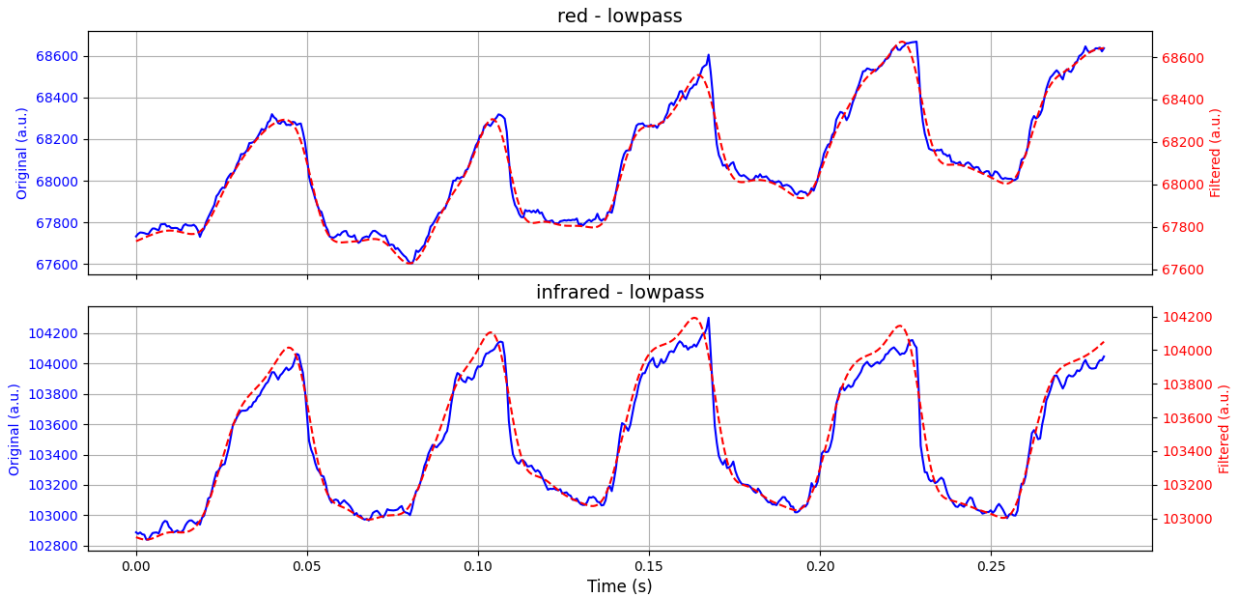


Fig. 2. Sensor data from the MAX30102 after applying a Butterworth low-pass filter.

F. Model Evaluation

Model performance was evaluated using standard evaluation metrics, including precision, recall, F1-score, and support for each class. These metrics provide a comprehensive understanding of the effectiveness of each model in distinguishing between classes.

To enhance the robustness of the model, 10-fold cross-validation was used. This technique partitions the dataset into 10 subsets, where each subset serves as a test set once, while the remaining subsets are used for training. This approach mitigates overfitting and ensures that the models generalize well to unseen data.

III. RESULTS AND DISCUSSIONS

Fig. 3 compares the signals from the MAX30102 and Shimmer3 sensors during hand movements. The first graph shows the infrared (IR) LED signal against the sEMG signal, while the second graph compares the red (RED) LED signal with the sEMG signal.

The comparison of the signals reveals some important insights. The observed changes in signal amplitude with each hand movement - opening and closing - demonstrate the sensors' ability to capture dynamic hand gestures effectively. The results suggest that the MAX30102 sensor has significant potential to measure muscle activity. This potential is supported by the similarity of the results obtained in previous studies [3] with comparable sensors.

However, to fully validate its effectiveness, it is essential to test it under various conditions. For example, future studies should examine sensor performance in scenarios involving muscle fatigue, skin perspiration, and different types of hand movements. These additional tests will help assess the sensor's robustness and reliability in real-world settings.

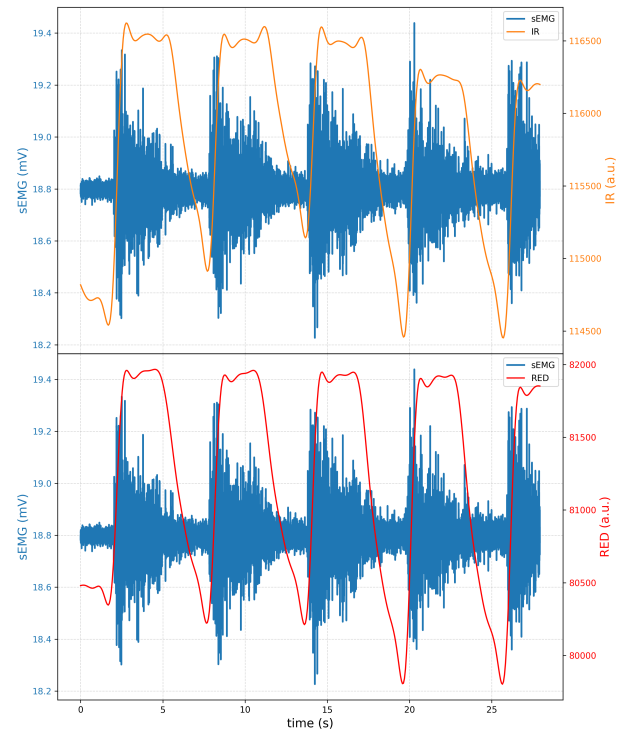


Fig. 3. Comparison of sEMG signals from Shimmer3 with red and infrared LED signals from MAX30102 during hand movements.

Fig. 4 displays the exploratory analysis of the red rolling mean feature to distinguish between target classes 0 and 1. In the histogram with kernel density estimation (KDE), the values for class 0 cluster around -1 , while those for class 1 lie near $+1$. The two distributions are well separated, with minimal overlap, indicating strong discriminative power for

this variable. The infrared feature yielded a similar result; therefore, the corresponding figure was omitted for brevity.

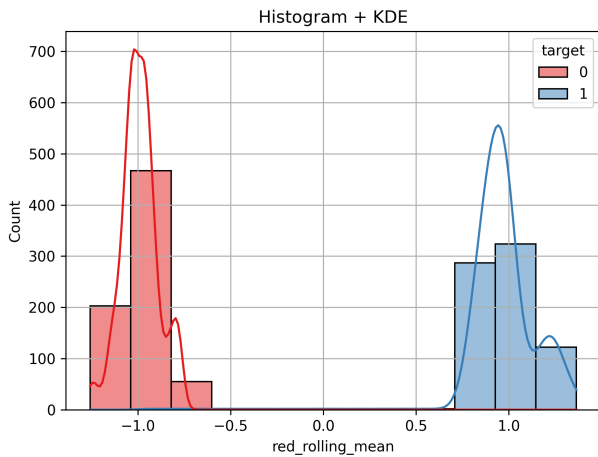


Fig. 4. Exploratory analysis of the red rolling mean feature to distinguish the target classes

Fig. 5 illustrates the accuracy of each model over ten different folds during cross-validation. This visualization highlights performance variability across different data folds. The Decision Tree and K-Nearest Neighbors models achieved notably higher accuracy compared to others.

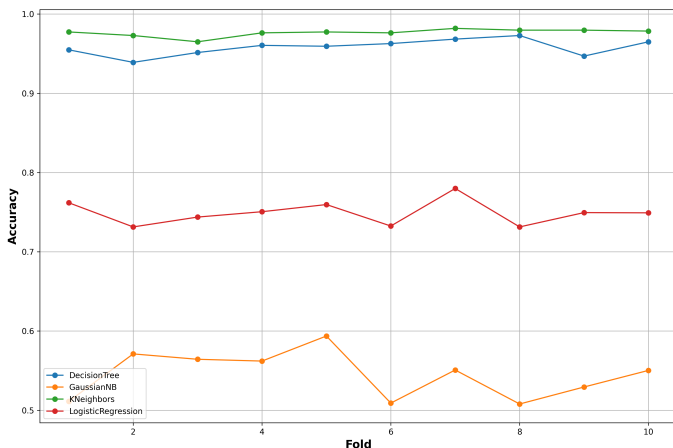


Fig. 5. Model accuracy across ten cross-validation folds.

Fig. 6 shows the mean accuracy of each model along with the standard deviation. The K-Nearest Neighbors model achieved the highest average accuracy of 98%. The Decision Tree model recorded an average accuracy of 96%, with a standard deviation of $\pm 1\%$.

IV. CONCLUSIONS AND FUTURE WORKS

The MAX30102 sensor demonstrated effective performance in detecting muscle activity and distinguishing between hand opening and closing movements. The sensor achieved high accuracy, with models such as K-Nearest Neighbors reaching 98% and Decision Tree achieving $96\% \pm 1\%$. These results

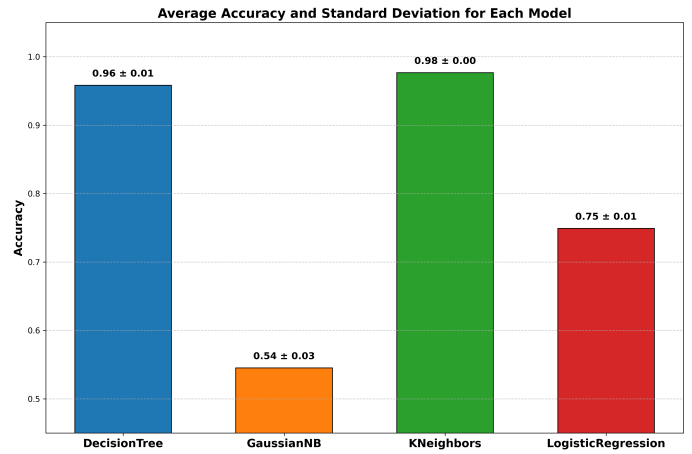


Fig. 6. Mean model accuracy with standard deviation.

validate the sensor's potential for accurately capturing and monitoring hand movements, highlighting its promise for applications in prosthetics.

Despite these promising results, the accuracy was achieved with a limited number of subjects, which can introduce a degree of bias. To ensure the sensor's robustness and generalizability, further validation with a more diverse group of participants is necessary, including individuals with varying skin tones, body types, and amputees.

Furthermore, future research should explore its ability to quantify the force exerted during these movements and consider integrating additional sensors. Such enhancements could broaden the sensor's applications and improve its effectiveness in more complex and real-world scenarios.

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